Cairo University
Faculty of Computers and Artificial Intelligence

CS251

Intro. to Software Engineering

Personal Budgeting Software

Software Requirements Specifications

Version 1.0

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CS251: Phase 1 – The Originals
Project: Personal Budgeting

Software Requirements Specifications

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CS251: Phase 1 – The Originals
Project: Personal Budgeting

Software Requirements Specifications

Team

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Document Purpose and Audience

Purpose

The purpose of this document is to standardize the software requirements agreed upon by all relevant stakeholders involved in this project. It aims to clearly define these requirements in an easy-to-understand format, ensuring consistency and clarity for all parties

Audience

- Software Developers & Engineers
- Investors & Stakeholders

Introduction

Software Purpose

This software focuses on simplifying budgeting for various expenses, such as daily necessities, bills, and entertainment, while also helping users save for future goals like travel, home purchases, or debt repayment. It provides tools to track spending, set savings targets, and maintain financial stability

Software Scope

The personal budgeting software helps users track income, expenses, and savings while setting and monitoring financial goals. It provides budget creation, spending insights, and automated tracking through bank integration. With secure data protection and multi-platform accessibility, it ensures efficient financial management

Software Requirements Specifications

Definitions, acronyms, and abbreviations

Phrase	Definitions
Budget	A financial plan that helps individuals track their income, expenses, and savings over a period
Expense	Money allocated by a person to spend on services
Income	The money earned from sources as salary, business and investments
Saving Goal	General purposes or some business or investments in any things and any financial specific target
API (Application Programming Interface)	A technology that allows different software applications to communicate with each other
E-Wallet (Electronic Wallet)	A digital payment system that stores funds for online and offline transactions.
Transaction	Any financial activity, such as deposits, withdrawals, or purchases.
UI/UX (User Interface / User Experience)	The design of application that can anyone use it in simple interface can any user to use this software or application
Budgeting App	A software application that helps users manage their personal finances by tracking income and expenses.
Recurring Expense	An expense that happens regularly, such as rent, subscriptions, or loan payments
Net Income	The amount of money a person earns after taxes and deductions.
Debt Repayment	The process of paying back borrowed money, such as loans or credit card balances



Software Requirements Specifications

Syncing	The process of automatically updating financial data between the software and external accounts

Requirements

Functional Requirements

Requirement ID	Requirement Details
FR-01	The application shall have user management like: • Users can register, log in, and log out securely • Users can update personal details and change passwords. • The system supports multiple user roles (basic user, premium user, admin).
FR-02	The application shall display a box to input **National ID** and **password** if the user chooses to log in, Below the input fields: • There shall be a **"Register"** button for new users to create an account. • A **"Forgot Password?"** link shall be available for users to recover their accounts. • The login box shall include a **"Remember Me"** checkbox for users who wish to stay signed in. • A **"Login"** button shall be provided to authenticate and access the system securely
FR-03	The application shall have display a dashboard if the user need to register, the dashboard consists of Full Name, Profile Picture (Optional), Email

Software Requirements Specifications

	Address, Phone Number, **National ID, Password**, default location and User Type (Basic, Premium, Admin)
FR-04	Users shall see a **financial summary** section, including **total income, total expenses, and remaining budget**
FR-05	The system shall display a **list of recent transactions** with category icons and include a **graph or pie chart** for **spending trends and savings progress**.
FR-06	Users shall have an option to **set and track budgets** for different categories, The budget screen shall display **current spending vs. budgeted amount** using progress bars and The application shall provide **alerts** when the user is **close to exceeding** their budget.
FR-07	The application shall have functionalities for Income & Expense Tracking like: • Users shall be able to **add, edit, or delete** income and expense transactions. • The system shall automatically categorize expenses and display them in a **categorized expense list** • Users shall have an option to **attach receipts** to transactions
FR-08	Users shall be able to create **financial goals** (e.g., saving for a car, vacation) and display a **progress bar** showing goal completion percentage.
FR-09	The application shall have to **view financial reports** in visual formats such as **graphs and charts** allow users to export reports in **PDF**, **Excel**, or **CSV** formats
FR-10	The application shall display notifications for **bill due dates** and **budget alerts** or **unusual spending behavior**
FR-11	Users shall be able to **update their profile information**, including name, email, and password they can modify their personal details in the settings section, the settings page shall provide an option to enable Two-Factor Authentication (2FA) with activate 2FA for additional security (via SMS or email) and users shall be able



Software Requirements Specifications

	to manage **linked bank accounts and e-wallets** from the settings and they can add, remove, or update linked financial accounts for transaction syncing

Non Functional Requirements

Measure	Details
Usability	The application shall support multiple languages, including **English and Arabic,** Ensures accessibility for diverse users. Language selection should be available in settings. compatible with **screen readers**, ensuring accessibility for visually impaired users and the user interface (UI) shall be **intuitive and easy to navigate**, requiring no more than **three clicks** to access key features, Navigation should be simple, ensuring users can quickly find budgeting, expenses, and reports.
Performance	• The application shall respond to user actions (e.g., login, transaction entry) within **2 seconds** • The system shall support up to **1,000 concurrent users** without performance issues • Financial reports shall be generated within **5 seconds**, even for **one year's data**
Reliability & Availability	• The system shall maintain **99.9% uptime**, limiting downtime to a maximum of **8.76 hours per year** • Automatic data backups shall occur every 24 hours to prevent data loss
Scalability & Maintainability	• The system shall be designed to scale and handle up to **10 times** the initial user base, Ensures infrastructure can support future growth without performance issues • A **modular software architecture** shall be used for easier maintenance and updates that allows developers to update individual modules without affecting the entire system.



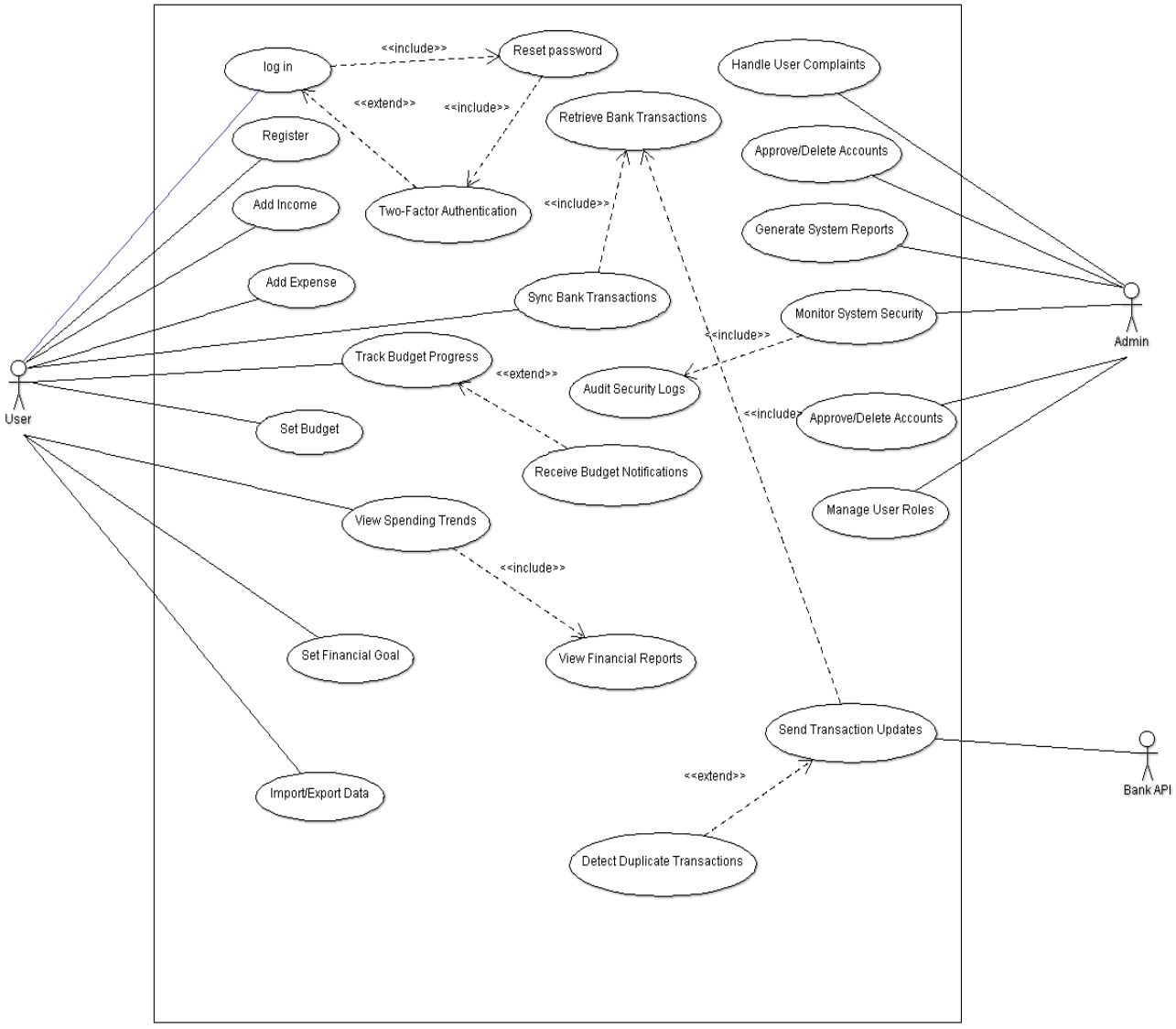
Software Requirements Specifications

	• System updates shall be deployed **seamlessly**, without causing service interruptions, updates should be rolled out in phases or during non-peak hours to prevent disruptions
Compatibility	The system shall be compatible with **Windows**, **macOS**, **Android**, **iOS** and web version shall support major browsers, including **Chrome**, **Firefox**, **Edge**, and **Safari.** The mobile application shall be responsive, supporting screen sizes as small as 5 inches. The design should be mobile-friendly with touch-optimized controls for easy navigation
Security & Data Protection	• The system shall use **AES-256 encryption** to protect sensitive data, financial data, login credentials, and personal details remain secure in storage and during transmission. • Multi-Factor Authentication (MFA) shall be required for account login, Users must verify their identity using an extra layer of security, such as an email or SMS code • User financial data shall remain private and will not be shared without explicit consent, they must be informed before any data is shared, and opt-in consent should be required for third-party integrations • The system shall **automatically log out** users after **10 minutes of inactivity,** prevents unauthorized access in case a user forgets to log out from a shared or public device.

Software Requirements Specifications

System Models

Use Case Model





Software Requirements Specifications

Enriched User Stories

- ### User Story #1

User Story ID	US #1
User Story Name	Registering a New Account
Actors	User
Description	**As** a new user I **like** to create an account in the system **So** I can access the system's features and manage my profile.
Per condition	
Post condition	
Acceptance Criteria	**Given** a user wants to register an account **When** they fill out the registration form with a username, email, and password **Then** the system should validate that the email format is correct

- ### Scenarios

Normal Scenario

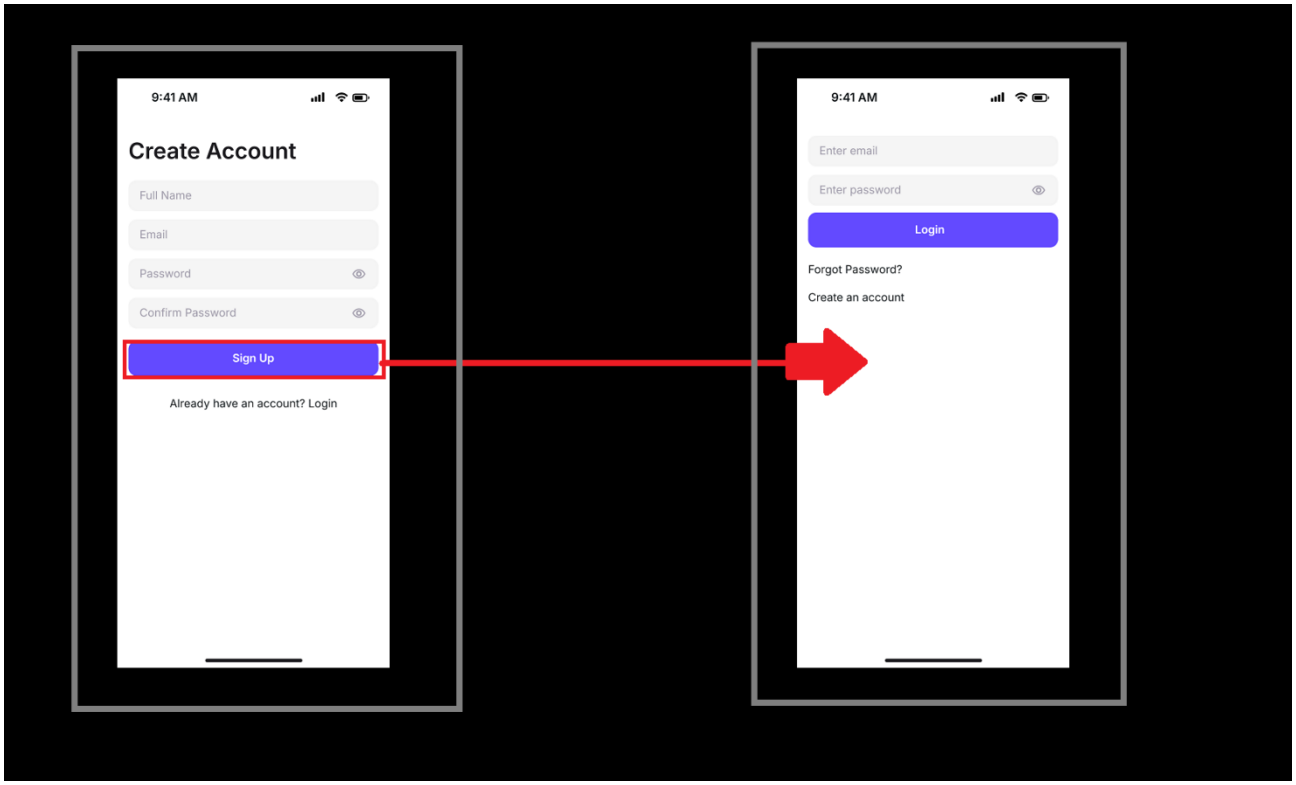
Actor Action	System Response
1- User clicks on **Sign Up**.	
	2- System displays the Sign Up page. 3- System displays fields to enter **Username, Email, and Password**.
4-User fills in the required fields.	
	5- System validates input. 6-System sends a confirmation email.
7- User verifies email.	
	8-System activates the account.
9-User logs in.	

Software Requirements Specifications

Exceptional Scenario

Actor Action	System Response
1- User enters an invalid email.	
	2- System shows "Invalid email format."

- ### Screen Design



- ### Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
username	String/50	Must be unique, no special characters except _ and .
email	String/100	Must be a valid email format (e.g., user@example.com)

Software Requirements Specifications

Element Label	Type/Length	Data Validation / Business Rule
password	String (Encrypted)	Min 8 characters, must include letters, numbers, and symbols
status	Enum (Pending, Active)	Default: Pending, changes to Active after email verification

---

- **User Story #2**

User Story ID	US #2
User Story Name	Logging into the System
Actors	User
Description	**As** registered user, I **like** log into the system using my credentials, **So** I can access my personal dashboard and system functionalities..
Per condition	
Post condition	
Acceptance Criteria	**Given** user wants to log in **When** they enter their username/email and password in the login form **Then** the system should authenticate the user and grant access if the credentials are correct

- **Scenarios**

  Normal Scenario

Actor Action	System Response
1- User clicks Login.	
	2- System shows login form.
3- User enters valid credentials.	

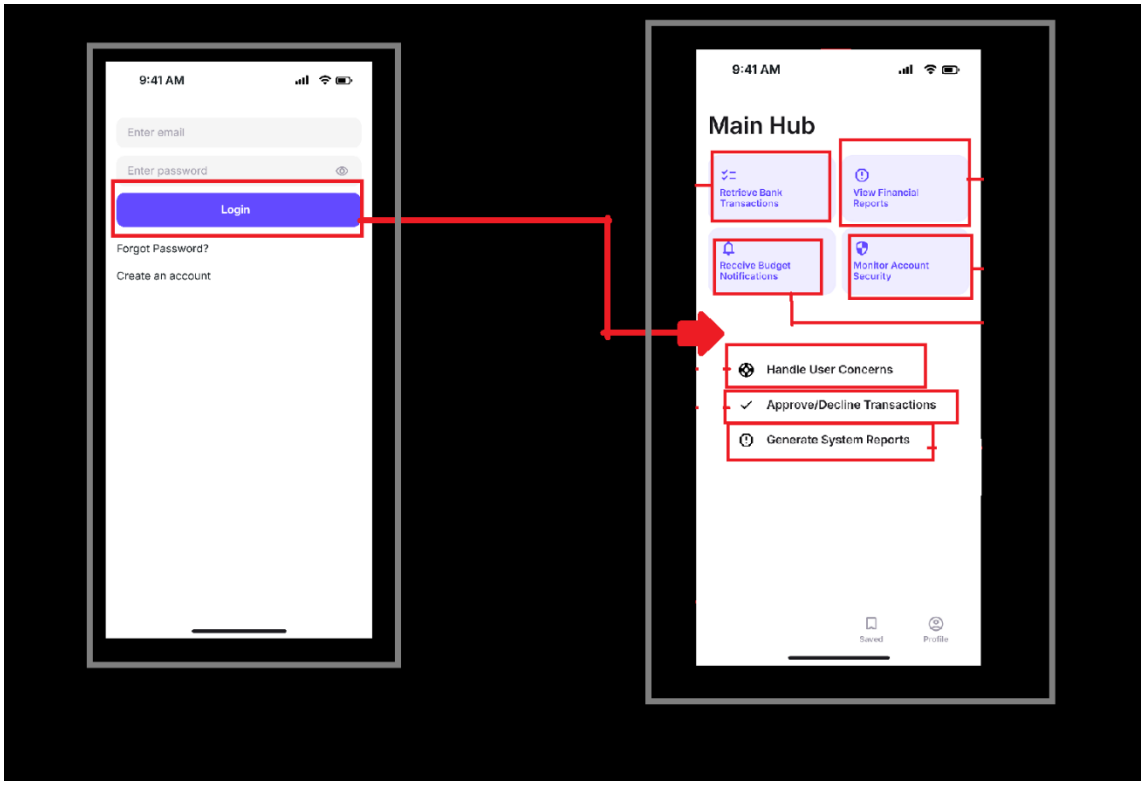
Software Requirements Specifications

	5- System verifies credentials. 6- User is granted access.

Exceptional Scenario

Actor Action	System Response
1- User enters incorrect credentials.	
	2- System shows an error message.

- **Screen Design**



Software Requirements Specifications

- **Data Dictionary:**

Element Label	Type/Length	Data Validation / Business Rule
username	String/50	Must exist in the system
Login Attempts	Integer	Max attempts before temporary lockout
password	String (Encrypted)	Must match the stored password
status	Enum (Pending, Active)	Default: Pending, changes to Active after email verification

---

- **User Story #3**

User Story ID	US #3
User Story Name	Setting Budget Alerts
Actors	User
Description	**A**s a user I **like** to set alerts for when my spending approaches my budget limit **so that I** can be notified before exceeding my financial plans.
Per condition	
Post condition	
Acceptance Criteria	**Given a** user wants to manage budget alerts **When** they enable or disable budget alerts and set a threshold percentage **Then** the system should store their preferences

- **Scenarios**

  Normal Scenario

Actor Action	System Response
1- User navigates to budget settings.	
	2- System displays alert configuration options.



Software Requirements Specifications

3- User enables alerts and sets a threshold (e.g., 80%).	
	4- System saves the settings.
5- User spends money, reaching 80% of the budget.	6- System sends a notification: "You have used 80% of your budget."

Exceptional Scenario

Actor Action	System Response
1- User sets an invalid threshold (e.g., 150%).	
	2- System displays an error message: "Threshold must be between 1% and 100%."
3- User disables alerts but expects notifications.	4- System does not send alerts since they are disabled.

Software Requirements Specifications

- **Screen Design**



- **Data Dictionary:**

Element Label	Type/Length	Data Validation / Business Rule
Alert Threshold	Integer (1-100)	Must be between 1% and 100%



Software Requirements Specifications

- ## User Story #4

User Story ID	US #4
User Story Name	Categorizing Expenses Within a Budget
Actors	User
Description	**As** a user, I **like to** categorize my expenses under different budget categories, **so** that I can better understand where my money is going.
Per condition	
Post condition	
Acceptance Criteria	**Given** a user wants to track expenses within budget categories **When** they assign an expense to a predefined budget category **Then** the system should categorize the expense automatically if possible

- ## Scenarios

Normal Scenario

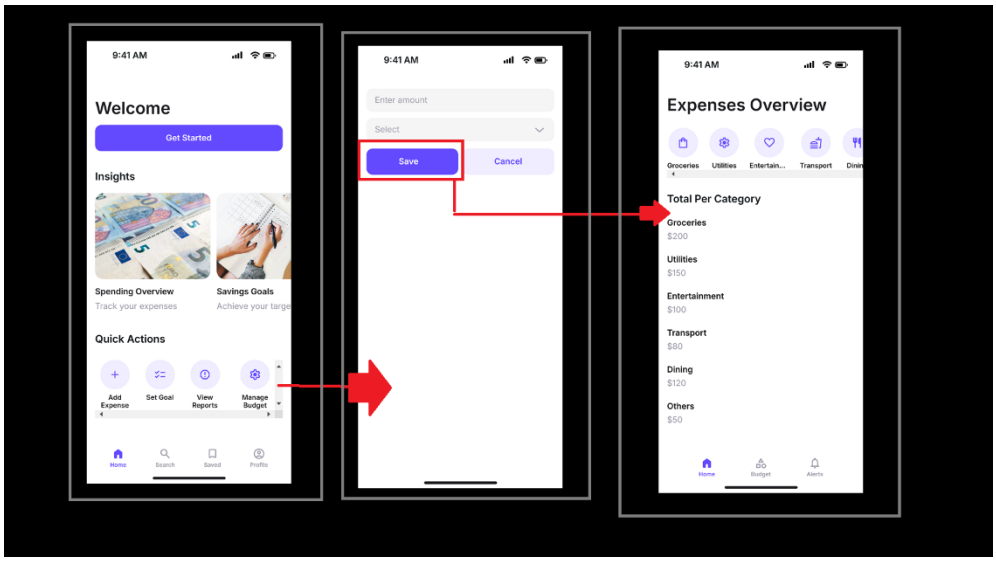
Actor Action	System Response
1- User navigates to the expense entry page.	
	2- System displays a form with an optional category field.
3- User selects a category (e.g., "Groceries").	
	4- System stores the expense under that category.
5- User views their budget summary.	6- System shows categorized spending breakdown.

Software Requirements Specifications

Exceptional Scenario

Actor Action	System Response
1- User leaves the category blank.	
3- User enters an invalid category..	2- System automatically assigns a default category (e.g., "Uncategorized"). 4- System displays an error: "Invalid category selected."

- ### Screen Design



- ### Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Expense Category	String (50)	Must match a predefined category
Expense Amount	Decimal (10,2)	Must be positive



Software Requirements Specifications

- ## User Story #5

User Story ID	US #5
User Story Name	Exporting Budget Reports
Actors	User
Description	**As a** user, I **like** to export my budget reports in different formats, **so** that I can share them or analyze them later.
Per condition	
Post condition	
Acceptance Criteria	**Given** a user wants to export a budget report **When** they choose a file format (PDF, CSV, or Excel) and request an export **Then** the system should generate a report including all budget details

- ## Scenarios

Normal Scenario

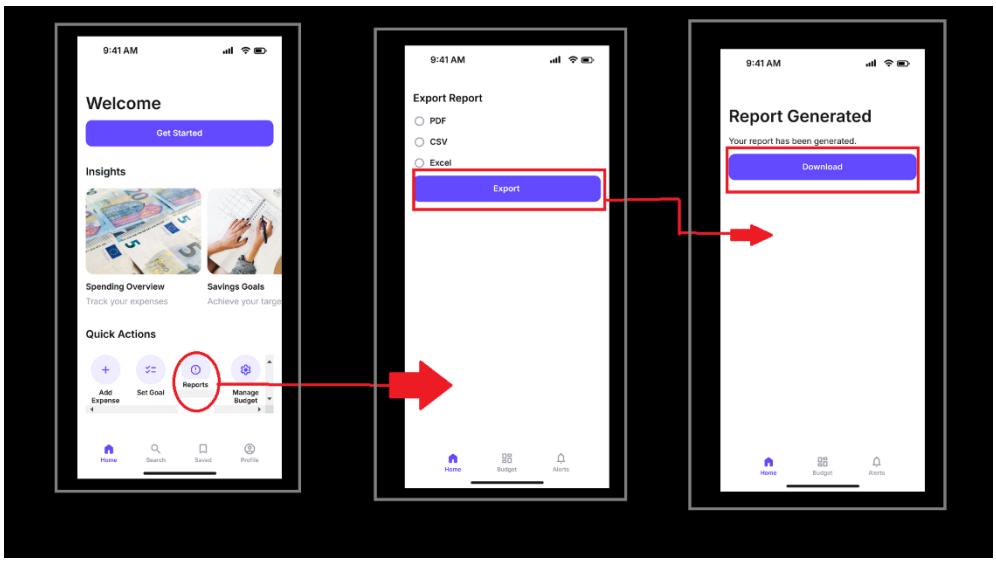
Actor Action	System Response
1- User navigates to the reports page.	
	2- System displays available report formats.
3- User selects PDF and clicks "Export."	
	4- System generates the report.
5- System displays a success message: "Your report has been generated."	
	6- User downloads the report.

Software Requirements Specifications

Exceptional Scenario

Actor Action	System Response
1- User tries to export without selecting a format.	
3- Report generation fails due to a server issue.	2- System prompts the user to choose a format. 4- System displays an error: "Failed to generate report. Try again later."

- ### Screen Design



- ### Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Report Format	String (PDF, CSV, Excel)	Must be a valid format



Software Requirements Specifications

- ## User Story #6

User Story ID	US #6
User Story Name	Viewing Monthly Spending Trends
Actors	User
Description	**As a** user, I **like to** see my monthly spending trends, **so** that I can understand my financial habits over time.
Per condition	
Post condition	
Acceptance Criteria	**Given** a user wants to analyze spending trends **When** they view the visual representation of their spending **Then** the system should display spending trends

- ## Scenarios

Normal Scenario

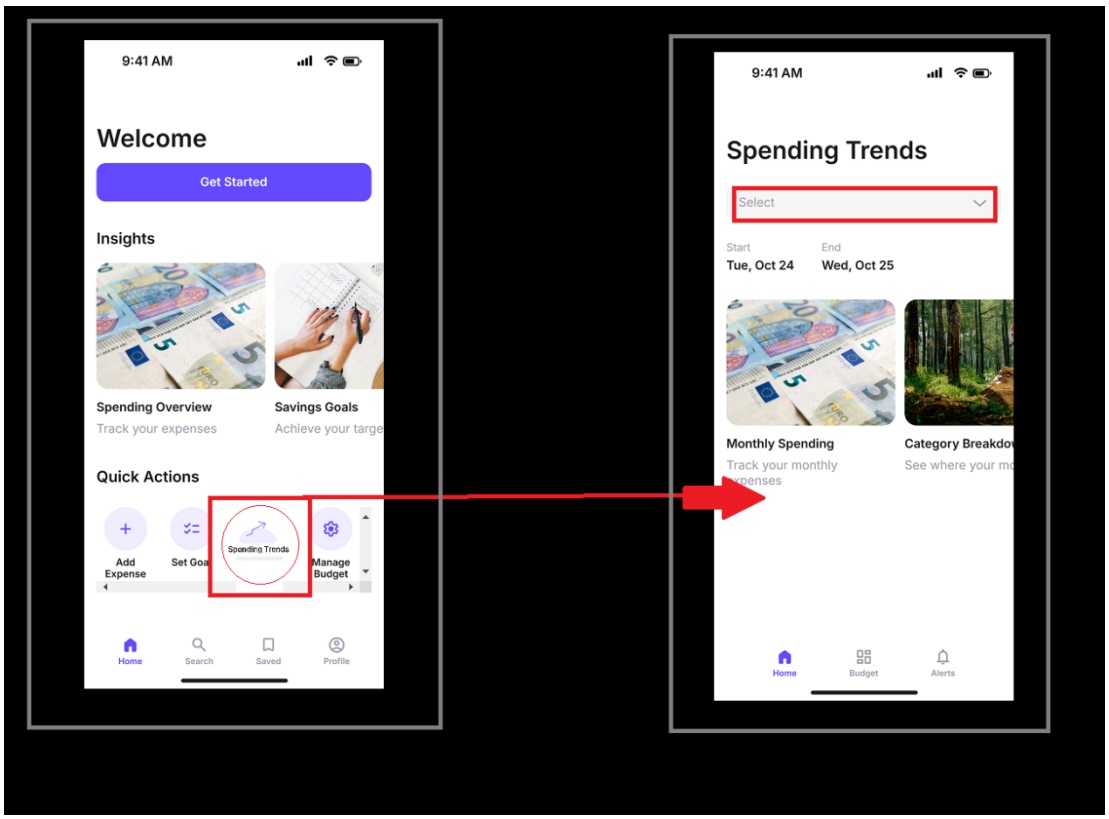
Actor Action	System Response
1- User navigates to the spending trends page.	
	2- System displays graphs showing monthly spending.
3- User filters by category (e.g., "Food").	
	4- System updates the graph to show food expenses only.
5- User compares spending from January to February.	
	6- System highlights increases or decreases.

Software Requirements Specifications

Exceptional Scenario

Actor Action	System Response
1- User selects a time period with no data	
3- User attempts to compare two identical months.	2- System displays "No data available for this period." 4- System displays a message: "Select different months for comparison."

- ### Screen Design





Software Requirements Specifications

- ## Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Time Period	String (YYYY-MM)	Must be a valid date format
Expense Type	String (50)	Must match predefined categories

- ## User Story #7

User Story ID	US #7
User Story Name	Syncing Bank Transactions to Budget
Actors	User
Description	**As a** user, I **like to** sync my bank transactions with my budget, **so** that I can automatically track my spending.
Per condition	
Post condition	
Acceptance Criteria	**Given** a user wants to connect their bank account securely **When** they link their bank account to the system **Then** the system should import recent transactions and categorize them automatically

- ## Scenarios

Normal Scenario

Actor Action	System Response
1- User navigates to the bank sync page.	
	2- System prompts for bank login details.
3- User authorizes the connection.	
	4- System imports recent transactions.
5- System categorizes transactions and displays them in the budget.	
	6- User reviews the imported data.

Software Requirements Specifications

Exceptional Scenario

Actor Action	System Response
1- User enters incorrect bank credentials.	
3- Bank service is down.	2- System displays an error: "Invalid login details." 4- System displays an error: "Unable to connect to bank. Try again later."

- ### Screen Design



- ### Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Bank Name	String (100)	Must be a valid date format
Account Number	String (20)	Must be a valid account number



Software Requirements Specifications

- ## User Story #8

User Story ID	US #8
User Story Name	Changing Currency Preference
Actors	User
Description	**As** a user, I **like to** select my preferred currency, **So** that I can view my budget in the currency I use.
Per condition	
Post condition	
Acceptance Criteria	**Given** a user wants to select a currency for financial values **When** they choose a currency from the dropdown menu **Then** the system should convert all financial values to the selected currency

- ## Scenarios

Normal Scenario

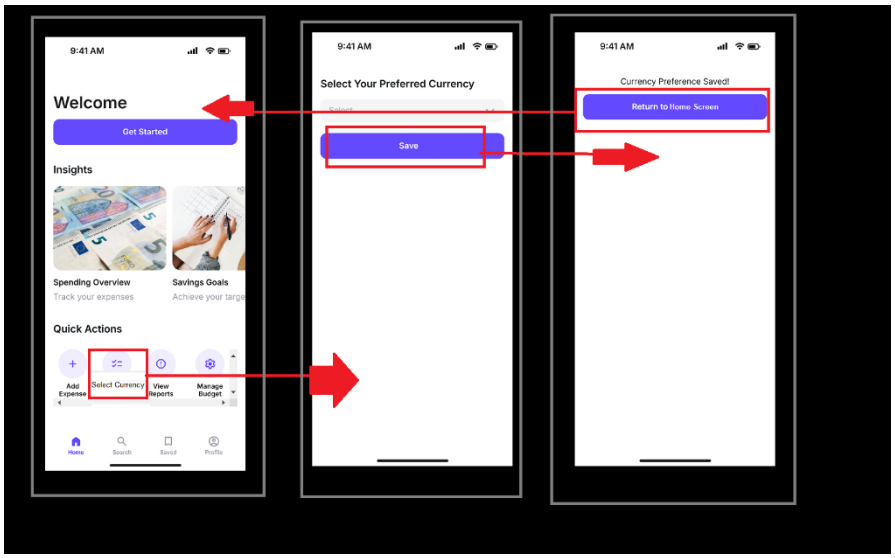
Actor Action	System Response
1- User navigates to settings.	
	2- System displays settings options.
3- User selects "Currency Preference."	
	4- System shows a dropdown with available currencies.
5- User selects a new currency (e.g., EUR).	
	6- System updates all financial values to the selected currency.
7- User saves changes.	
	8- System stores the new currency preference and displays a confirmation message.

Software Requirements Specifications

Exceptional Scenario

Actor Action	System Response
1- User selects an unsupported currency.	
3- User closes the settings without saving.	2- System displays an error: "This currency is not supported." 4- System discards changes and keeps the previous currency.

- ### Screen Design



- ### Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Currency Code	String (3)	Must be a valid currency code (e.g., USD, EUR, GBP).
Exchange Rate	Decimal (10,4)	Must be a positive value.



Software Requirements Specifications

- ## User Story #9

User Story ID	US #9
User Story Name	Logging Out of the System
Actors	User
Description	**As a** user, I **like to** log out of my account, **So** that I can keep my financial information secure**.**
Per condition	
Post condition	
Acceptance Criteria	**Given** a user is logged into the system **When** they click the "Logout" button in the settings menu **Then** the system should log them out and return to the login screen

- ## Scenarios

Normal Scenario

Actor Action	System Response
1- User clicks on the profile icon.	
	2- System displays a dropdown menu.
3- User selects "Logout."	
	4- System logs the user out and redirects to the login page.
5- User is logged out successfully.	6- System clears session data.

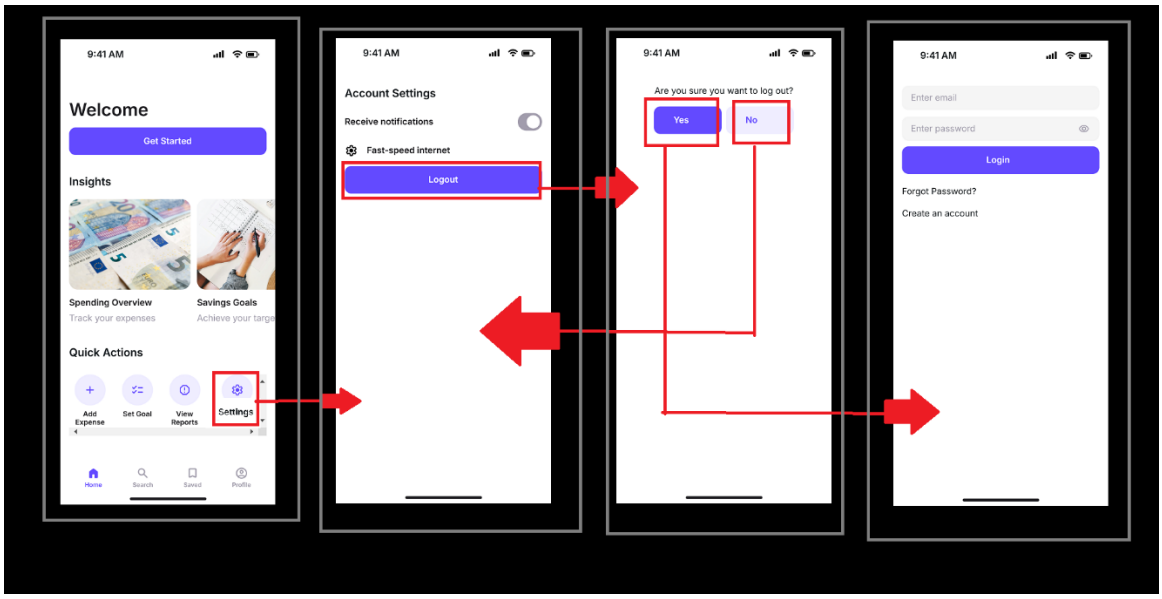
Exceptional Scenario

Actor Action	System Response

Software Requirements Specifications

1- User tries to log out but has an unstable internet connection.	
3- User's session expires due to inactivity.	2- System displays an error: "Network error. Try again later." 4- System automatically logs the user out and redirects to the login page.

- ## Screen Design

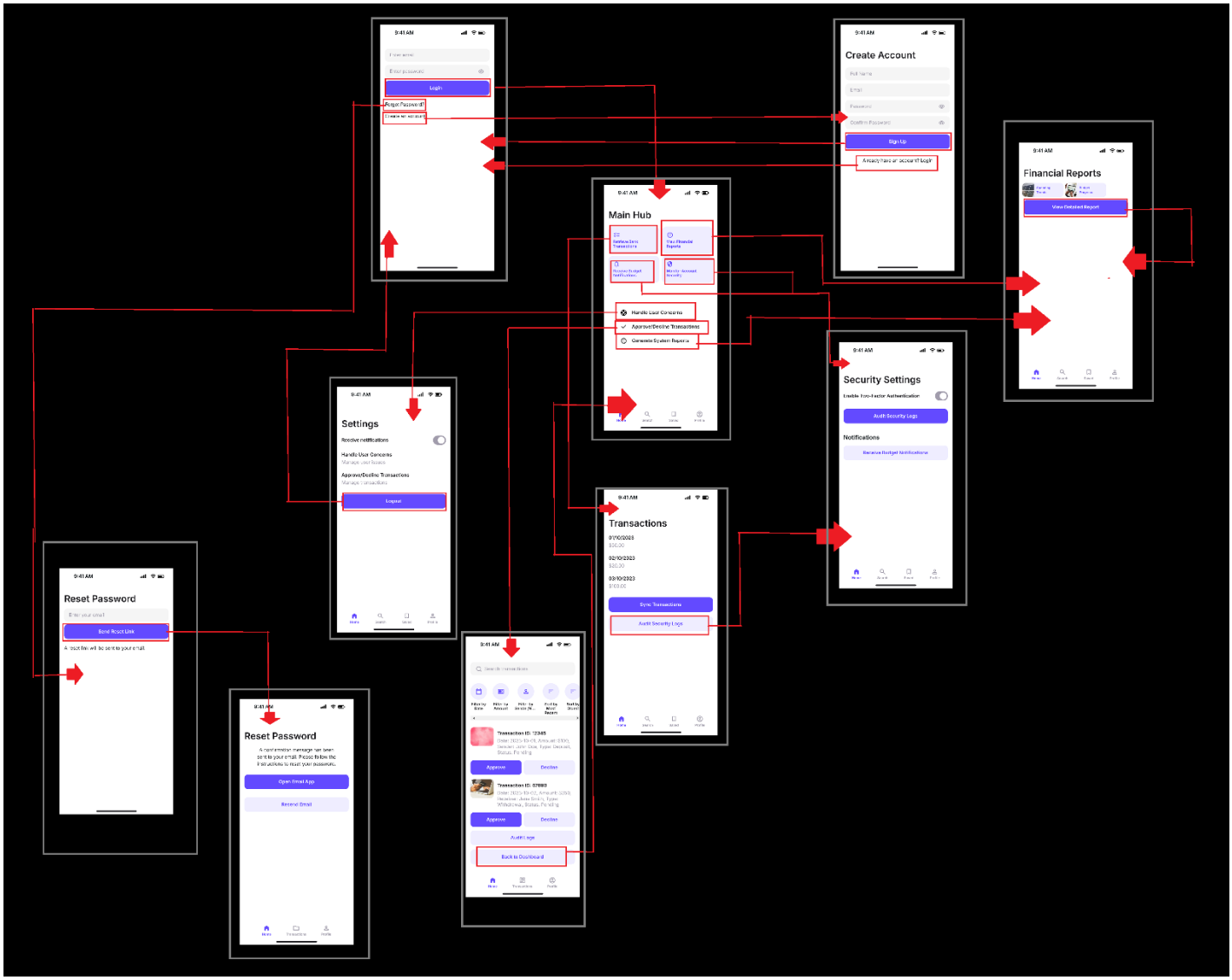


- ## Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
User ID	String (3)	Must be a registered user.
Exchange Rate	String (256)	Must be valid and active.

Software Requirements Specifications

System Navigation Map



Software Requirements Specifications

Tools

- **ArgoUML for use case**
- **Whimsical, Painter and Fiegma for all designs in the report**

Ownership Report

-

Student	Items he created
Abdallah Kareem Abdallah	• Functional Requirements • Use Case Diagram • Non-Functional Requirements • Part of Definitions
Reyad Youssef Ebrahim	• System Navigation Map • All of user stories • All Design of the System
Hady Hassan Ahmed	• Part of Definitions • Purpose • Scope • Document Purpose and Audience